

Jayadev Ghanta

(408) 497-0954 | jayadevgh@gmail.com | [jayadevgh.github.io](https://github.com/jayadevgh) | [linkedin.com/in/jayadev-gh/](https://www.linkedin.com/in/jayadev-gh/) | github.com/jayadevgh
Paul G. Allen School of Computer Science & Engineering, University of Washington

EDUCATION

University of Washington, Seattle

Expected June 2028

B.S. Computer Science and B.S. Mathematics (GPA 3.82/4.0)

- Coursework: Machine Learning, Distributed Systems, Intelligent Control, Systems Programming, Hardware/Software Interfaces, Data Structures & Parallelism, Probability, Real Analysis

Euler Circle (Advanced Math Institute)

2024 – 2025

Advanced pre-college coursework in Nonstandard Analysis, Cryptography, Combinatorial Game Theory, Complex Analysis

- Authored research in Ramsey Theory ([paper](#)), Elliptic Curve Cryptography ([paper](#)), and Error-Correcting Codes ([paper](#)).

EXPERIENCE

Cerebras Systems

May 2026 – Present

AI Engineering Intern

Sunnyvale, CA

- Worked with the kernel and model bring-up team on ML systems infrastructure for deploying transformer models

University of Washington — [Noah's ARK](#) (NLP Group)

2025 – Present

Undergraduate Researcher (1 of 10 undergrads)

Seattle, WA

- Engineered an end-to-end multimodal pipeline to construct a large-scale dataset for musical understanding research, linking 1000+ annotated scores from the Digital Library of Classical Music with 1M+ online performances.
- Developed conservative verification and curation systems using DTW-based alignment, learned audio embeddings, and transcription models to validate 200K+ YouTube performances against reference scores.

Dartmouth-Hitchcock Medical Center ([Levy Lab](#))

Summers 2023 – 2024

Research Intern (1 of 60 students selected)

Lebanon, NH

- Replaced costly destructive staining by training a GAN + GNN model that predicts immune-cell locations from standard H&E slides, validated on 4000+ images.
- Developed an interpretable weakly supervised Multiple Instance Learning (MIL) system to rank malignant cells in weakly labeled cytology slides, achieving 0.91 AUC on bladder cancer detection and highlighting cell-level diagnostic evidence.

[Independent Research](#) — Harvard (Dr. Akshay Jagadeesh)

2023 – 2024

Student Researcher

Remote

- Modeled continuous Alzheimer's progression under severe data scarcity by training conditional and Wasserstein GANs to generate 5000+ synthetic MRI scans across disease stages.

AWARDS

2026 Hack to the Future – 1st Place Track Winner ([Link](#))

USACO Gold Division 2026 | [Codeforces](#) Expert (1605)

2025 DubHacks – Top 3 Winner, AWS Track ([Link](#))

2x AIME Qualifier (Top 5%) 2022, 2023

PROJECTS

[Transformer Systems From Scratch](#) | *PyTorch, NumPy, LLMs*

2023 – 2025

- Trained two 12M-parameter decoder-only Transformers (character and BPE tokenization) and used average negative log-likelihood as a fluency score, achieving 93% accuracy on minimally corrupted sentence pairs.
- Implemented a full BPE tokenizer, [Transformer stack](#), and [NumPy-based deep learning microframework](#) from scratch, including tensors, autodiff, backpropagation, and optimizers.
- Matched nanoGPT perplexity and published 3 technical blogs on [Medium](#) read by 1000+ viewers on model internals and autodiff design.

[KeyGuard](#) | *Python, FastAPI, React, TypeScript, PyTorch, SQLite*

2026

- Developed a continuous authentication system using keystroke dynamics to detect session hijacking after login.
- Designed an embedding model with Mahalanobis-distance scoring, achieving 96.9% Recall@1 on a custom 10+ user dataset.

[Bond](#) | *Kotlin, Android Studio, Firebase, AWS Bedrock, BLE*

2025

- Created an Android app that detects nearby users in real time using Bluetooth Low Energy, enabling proximity matching.
- Implemented a cloud-backed user matching and tracking system using Firebase and AWS Bedrock.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Kotlin, JavaScript, SQL, LaTeX, Bash, HTML, CSS

Frameworks & Libraries: PyTorch, NumPy, Pandas, Scikit-learn, Hugging Face, Jetpack Compose, Firebase, AWS Bedrock, LangChain, Android BLE, Flask, React

Tools: Git, GitHub, VS Code, Android Studio, Google Cloud Run, Codex, Docker, Jupyter, Vim, Linux

Topics: Machine Learning, Deep Learning, NLP, ML Systems, Distributed Systems, Optimization, Cryptography, Statistics, Real Analysis